

Innovative Breast Cancer Diagnosis Model Based on Quantum Transfer Learning

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Abstract

Breast cancer remains one of the most prevalent cancers among women worldwide and causes high mortality. Early and accurate diagnosis is crucial to improving treatment outcomes and reducing mortality. This study introduces an innovative hybrid model. The model combines ResNet-50, EfficientNet-B0, and Vision Transformer (ViT), and applies advanced quantum transfer learning techniques to leverage their advantages in feature extraction, overcoming the obstacles of insufficient data, and further improving the model's performance. Experimental results demonstrate that the proposed model achieved 99.1% accuracy, 99.0% precision, 98.8% recall, and 98.9% F1 score. Those findings demonstrate that the proposed model achieved superior performance in the task of breast cancer diagnosis. This work provides a prospective study for efficient, accurate, and scalable breast cancer diagnosis solutions based on feature-level ensembling and quantum transfer learning in clinical settings.

CCS Concepts

• **Computing methodologies** → Artificial intelligence; Computer vision; Computer vision tasks; Visual inspection.

Keywords

Quantum Machine Learning, Ensemble Learning, Transfer Learning, Medical classification, Breast Cancer

ACM Reference Format:

Xinyue Yu. 2025. Innovative Breast Cancer Diagnosis Model Based on Quantum Transfer Learning. In *2025 International Conference on Artificial Intelligence and Computational Intelligence (AICI 2025), February 14–16, 2025, Kuala Lumpur, Malaysia*. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3730436.3730453>

1 INTRODUCTION

As one of the major threats to women's health, the diagnosis and treatment of breast cancer has always been the focus of medical research. Data show that breast cancer is the major cause of cancer-related death among women worldwide [1]. Therefore, improving

the early diagnosis rate of breast cancer is critical in reducing the mortality rate. This would significantly enhance the treatment success rate and reduce the financial burden on patients. Currently, breast cancer diagnosis mainly relies on imaging examinations such as mammography, ultrasound, and magnetic resonance imaging (MRI). These technologies enable clinicians to identify abnormal features within the breast that cannot be directly observed with the naked eye.

However, current diagnostic approaches face certain limitations. These methods typically require experts to observe and judge, which indicates a high cost. Furthermore, it is not feasible in undeveloped countries or regions where medical staff with limited medical personnel and scarce resources. Therefore, developers have been seeking more efficient and lower-cost diagnostic technologies. In recent years, with the rapid development of artificial intelligence technology, the application of machine learning in medical image analysis and diagnosis has made significant research progress and has been widespread adoption [2]. For example, Qasem Abu et al. [3] developed an accurate and comprehensive breast cancer diagnosis framework by using ResNet-50, which is a pre-trained convolutional neural network on ImageNet, to classify histopathological microscope images. Their work successfully distinguished between benign and malignant tumors. Maheshvar Chandrasekar et al. [4] explored the application and performance of the EfficientNetB3 architecture in the breast cancer classification tasks. By comparing it with other architectures such as DenseNet and ResNet, EfficientNetB3 showed faster convergence and higher classification performance in the task of classifying breast histopathological images from the BreakHis dataset. In addition, recent studies have shown that machine learning can provide personalized recommendations for patients during diagnosis and treatment, ultimately improving patient treatment outcomes [5]. Machine learning-related approaches show considerable potential for application in precision medicine, such as the prediction of the growth of tumors or identification of biomarkers [6].

Although machine learning technology has achieved remarkable results in image classification tasks, it still faces significant challenges. For instance, medical solutions based on machine learning usually face critical problems such as a high dependence on training data quality, insufficient generalization ability across data sets, and interoperability [7]. Compared with other types of lesions, machine learning solutions in breast cancer diagnosis present greater challenges: including the lack of comprehensive, balanced, and well-annotated datasets, class imbalance, and model performance limitations due to label noise [8]. These factors require the model

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AICI 2025, Kuala Lumpur, Malaysia

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ACM ISBN 979-8-4007-1363-7/2025/02

<https://doi.org/10.1145/3730436.3730453>

to be highly flexible and adaptable, as traditional models often fail to achieve.

In response to these challenges, this study combined several latest technologies to preliminarily design and implement an efficient and accurate breast cancer diagnosis model. The model employs Feature-Level Ensembling, combining the advantages of ResNet-50, EfficientNet-B0, and Vision Transformer (ViT), enabling a comprehensive understanding of image features. Meanwhile, quantum transfer learning based on Classic to Quantum (CQ) is further introduced. It combines the stability of classic machine learning models with the efficient feature extraction capabilities of the quantum layers. The classification accuracy and diagnostic performance of the model are further improved based on the quantum layer, which fully demonstrates the potential of quantum transfer learning in breast cancer diagnosis. This model serves as a reliable tool in early diagnosis to ensure that it meets the demands of clinical scenarios. Meanwhile, it also provides a promising solution for overcoming the problem of breast cancer image datasets: small scale, high annotation difficulty, and imbalanced sample categories.

2 RESEARCH BACKGROUND AND RELATED WORK

2.1 Multi-model Ensemble Method

Multi-model ensemble is a technique in machine learning. It can combine the features of multiple models to improve the performance of the hybrid model [9]. In contemporary machine learning practice, a single model would be constrained by limitations. However, by employing a multi-model ensemble, these limitations can be mitigated as the combined strengths of models compensate for individual shortcomings. This approach is widely used in medical-related research because it has proven to be highly effective in enhancing model performance and generalization ability. For example, one study proposed a methodology to integrate multiple distinct models in the case of predicting cancer. The results showed that compared with other methods (such as using a single model), their model has significantly improved the prediction accuracy, providing an effective new approach for cancer prediction [10]. Another study achieved the high-precision classification of gross tumor volume (GTV), brainstem, and normal brain tissue based on radiomic features through multi-model ensembling, leveraging the advantages of various machine learning algorithms [11].

Feature-Level Ensembling is a technique of fusing models at the feature level. The features extracted by the models are gathered and fused, then passed to additional network layers for further processing. This approach can enhance feature representation by leveraging the diversity inherent in multiple models. Feature-Level Ensembling has found extensive applications in medical image recognition. For instance, a study has applied similar techniques to determine whether a lung lesion is caused by COVID-19. This saves labor costs, which means medical assistance can be provided to more patients under limited medical resources during the COVID-19 pandemic [12]. Another study proposed a novel model for automatically detecting and classifying different types of brain tumors. This model integrates the features extracted from three distinct basic models, and the final accuracy rate is as high as 98.37% [13]. These examples prove that Feature-Level Ensembling could bring

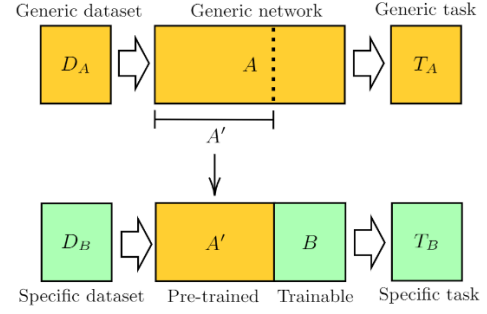


Figure 1: General representation of the transfer learning [19].

higher accuracy and model performance than adopting a single model.

2.2 Quantum Machine Learning and Transfer Learning

To further solve this problem and reduce the model's dependence on the quality and scale of the dataset, transfer learning is a method that is particularly suitable for this field. Transfer learning is a method that allows the model to apply the knowledge learned from one task to another related task, thereby improving the learning efficiency and performance. By leveraging pre-trained models, transfer learning can quickly train models with a dataset with a smaller scale and reduce time consumption [14]. In the field of medicine, transfer learning also has a wide range of applications and has developed rapidly in recent years [15]. It is used in many fields such as Alzheimer's disease, lung cancer, dentistry, etc [16], [17], [18]. A general representation of transfer learning can be expressed as Figure 1

Recent research has introduced an innovative approach to medical image classification and diagnosis: they proposed a method that integrates quantum machine learning with classical transfer learning to more efficiently transform the pre-trained model into a model that focuses on a specific problem [19]. There are three variants of transfer learning that are currently considered in the industry: classical to quantum (CQ), quantum to classical (QC) and quantum to quantum (QQ). After comparing these three variants, CQ has proven to be the most efficient variant and can produce the highest model performance. Another related study also proved that the CQ variant is superior to traditional transfer learning method on several typical datasets [20].

However, current academic articles also point out that the application of transfer learning in the field of breast cancer detection is still insufficient [8]; as well as in the field of quantum transfer learning. This paper actively considers the application of quantum transfer learning to solve the problems currently encountered in the field of breast cancer diagnosis and ultimately fill this research gap.

Furthermore, it is important to emphasize that the current academic research on quantum computing, especially in the quantum machine learning field, is still in its early stages of development. The hardware performance and availability of quantum computer

devices are not yet mature. The restricted number of available qubits constrains the operating efficiency of quantum machine learning models [21]. Therefore, although the current method theoretically combines the stability of classical transfer learning and the efficiency of quantum machine learning, its ideal performance needs to rely on more powerful quantum hardware in the future to be fully realized. At this stage, to explore the feasibility of this method, researchers can only use quantum computing simulators such as PennyLane for experiments and verification. They simulate quantum calculations on classical computers, and cannot fully demonstrate the potential performance of quantum computing. Nevertheless, their work lays the foundation for quantum transfer learning in breast cancer diagnosis.

3 PROPOSED WORK

This study proposed and developed a model that integrates a hybrid model framework. It uses feature-level ensembling and explores quantum transfer learning.

Firstly, this study employs Feature-Level Ensembling to stack ResNet-50, EfficientNet-B0, and ViT models to build the initial component of the hybrid model. This choice leverages the complementary strengths of the three models in feature extraction. ViT relies on the advanced attention mechanism to quickly grasp the global features of the image; however, one of the limitations of ViT is that it relies heavily on a large number of datasets [22]. Datasets often present a significant challenge in machine learning research in the medical field. Labeling an accurate medical dataset is extremely complex and costly [23]. Due to ethical and commercial restrictions, a lack of professional annotation, and other reasons, medical-related machine learning often cannot obtain large-scale data sets for training purposes [24]. The advantages of ViT may not be fully realized under this situation. Therefore, this study suggests introducing other models to compensate for this deficiency. ResNet-50 is based on a convolutional neural network. Despite its lower efficiency, it performs well when the dataset is relatively small [25]. Meanwhile, ResNet-50 performs well in extracting local features [26]. This synergy allows ResNet-50 to capture the local features ViT may overlook, ultimately enhancing the overall feature representation. Furthermore, EfficientNet-B0 employs an efficient compound scaling method, which has already shown great performance in studies related to breast cancer datasets [4]. EfficientNet-B0 can further enhance the efficiency and robustness of the hybrid model, making it particularly well-suited for the transfer learning approach which will be discussed in the subsequent part [27], [28].

Secondly, building on the work of Mari et al., this study incorporates quantum transfer learning in the hybrid model to enhance performance [19]. They proposed an architecture based on the CQ variant of transfer learning. It extracts features through pre-trained classical machine learning models but applies quantum layers for subsequent processing. This approach fully leverages the stability of traditional models and the feature extraction capabilities of quantum machine learning, enabling effective computation even in the situation of limited qubits. This can further mitigate the impact of insufficient datasets and limited annotation standardization. In conclusion, the integration of quantum transfer learning provides

a promising and innovative solution to breast cancer diagnosis. Figure 2 shows the Architecture of the proposed model.

4 EXPERIMENTAL AND RESULTS

4.1 Dataset

The Breast Cancer Histopathological Database (BreakHis) dataset consists of 9,109 microscopic images of breast tumor tissues, including 2,480 benign samples and 5,429 malignant samples. In histopathology, benign tumors are localized, grow slowly, and pose little threat to the body. In contrast, malignant tumors, i.e. cancer, are invasive and may lead to serious consequences or even death. These images were collected from 82 patients with 4 different magnifications. Each image has a resolution of 700x460 pixels, 3-channel RGB color, 8-bit depth, and is stored in PNG format. The tissue samples in the dataset were collected by a method called SOB, also known as partial mastectomy or excisional biopsy. Figure 3 presents a sample of the dataset.

4.2 Training

The dataset underwent a standardized image preprocessing pipeline before training, including resizing the image to 224x224 pixels, converting it to tensor format, and normalizing. This study also used data augmentation, which includes random horizontal flipping, vertical flipping, cropping, etc. This will increase the diversity of the training data by generating diverse transformation samples, thereby helping the learning of the model. Data augmentation was applied only to the training data, to ensure the stability of the validation dataset and its representation of real-world scenarios.

Breast cancer datasets, including the BreakHis dataset, usually have sample imbalance problems. Therefore, class balance is a critical issue when splitting the training and validation sets. Firstly, stratified sampling was adopted because it ensured that the proportion of samples in the training and validation sets was the same, thereby mitigating the impact of class imbalance on model performance. It is able to improve the model's performance on under-represented classes and provide representative validation data for evaluating the model's practical diagnostic ability. Secondly, this study integrated class weights into the loss function. It allows the model to focus more on the minority class samples during training epochs to address the issue of class imbalance. The weights were calculated by using a square root method. It scaled the class weights by their square roots to reduce the disparity between different classes. By this method, training instability caused by large differences in weights would be prevented. The model could more effectively identify minority class features thereby improving overall performance. This approach contributes to the model's robustness and reliability in practical breast cancer diagnostic tasks.

For the learning rate strategy, the Adam optimizer with L2 regularization was adopted to control overfitting. The Adam optimizer is an adaptive optimization algorithm. It dynamically applies an appropriate step size for each parameter, thereby improving the efficiency and stability of the training process. It also effectively avoids problems like exploding or vanishing gradients. L2 regularization can constrain the absolute values of the model parameters. It encourages the model to learn smoother features and reduces the

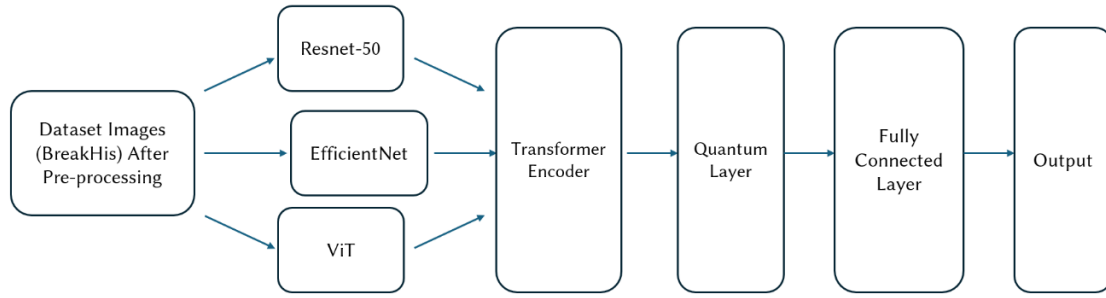


Figure 2: The architecture of the proposed model.

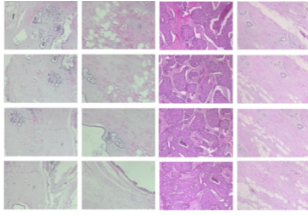


Figure 3: BreakHis Dataset samples.

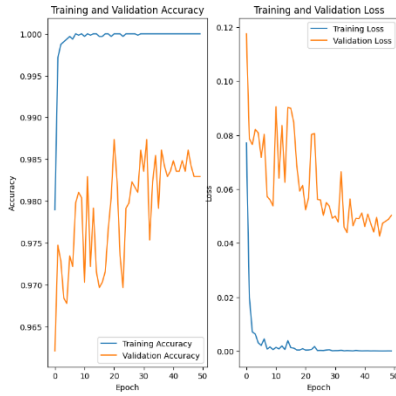


Figure 4: Learning curves for the ResNet-50 Model.

risk of overfitting. Additionally, this study employs a cosine annealing scheduler. It can gradually reduce the learning rate according to a cosine function during the training epoch. This approach brings a faster convergence in the early epochs with a larger learning rate; and a gradual decrease in the learning rate in the later epochs, enabling the model to tune finely and avoid overfitting.

4.3 Evaluation and Discussion

This study considered four models: ResNet-50, EfficientNet-B0, ViT, and the proposed model. This study conducted 50 training epochs with a batch size of 16 to evaluate each model's performance. The learning rate varied according to the characteristics of different models. Figures 4, 5, 6, and 7 show the learning curves of these four models.

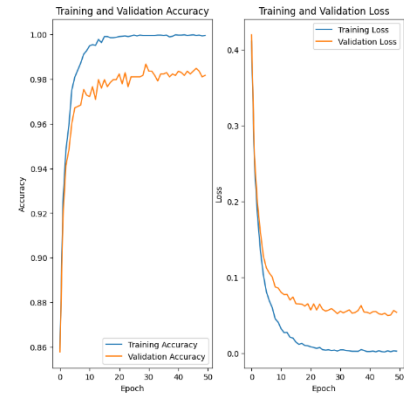


Figure 5: Learning curves for the EfficientNet-B0 Model.

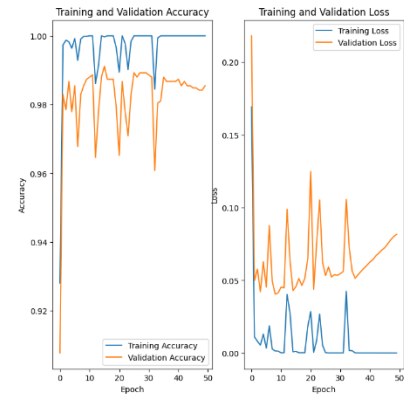


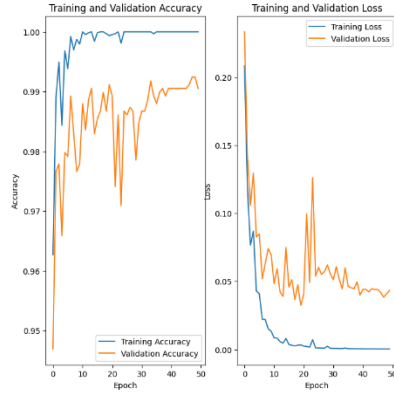
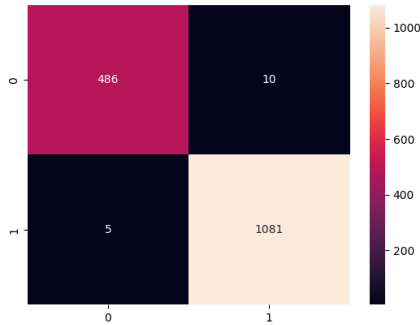
Figure 6: Learning curves for the ViT Model.

Also, this study comprehensively evaluates the performance of the selected models through a variety of performance indicators (Accuracy, Precision, Recall, F1-score). Those data are given in Table 1.

The results show that the proposed model outperforms other selected models across various performance metrics. The confusion matrix of the proposed model in Figure 8 is also provided to offer a more detailed performance evaluation. It indicates that the proposed model performs impressively in the breast cancer

Table 1: The comparison results between the Proposed Model and ResNet-50, EfficientNet-B0, and ViT.

	Accuracy	Precision	Recall	F1-score
	$ACC = (TP + TN) / (TP + TN + FP + FN)$	$PPV = TP / (TP + FP)$	$REC = TP / (TP + FN)$	$F1 = 2TP / (2TP + FP + FN)$
ResNet-50	0.983	0.984	0.977	0.980
EfficientNet-B0	0.982	0.980	0.977	0.979
ViT	0.985	0.983	0.983	0.983
Proposed Model	0.991	0.990	0.988	0.989

**Figure 7: Learning curves for the Proposed Model.****Figure 8: Confusion matrix for the proposed model.**

diagnosis tasks. The proposed model leverages the advantages of feature-level ensembling and quantum transfer learning. This enables the proposed model to improve its predictive performance for unfamiliar data, extract more discriminative features from datasets, and mitigate issues that may lead to biases in medical datasets and real-world diagnosis tasks.

5 CONCLUSION AND FUTURE WORKS

This study proposed a breast cancer diagnostic model based on Feature-Level Ensembling and Quantum Transfer Learning. The proposed model achieves exceptional diagnostic performance, as shown by experimental results on the BreakHis dataset. It offers an efficient and reliable solution for real-world breast cancer diagnosis

tasks. This study also established a link between quantum machine learning and clinical practice and demonstrated its further potential.

This work also highlights a limitation of quantum machine learning. For now, quantum computing hardware restricts the performance of quantum machine learning models, because of the limited number of available qubits and the high requirement for computing resources. Future research could leverage the next-gen quantum computing hardware when available and further optimize the proposed model, to make it more efficient, accurate, and economical.

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